Trey Kirscht

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Summary

Electrical Engineering Co-op student focused on embedded systems, FPGA development, and mixed-signal hardware. Built and bench-validated a 48-electrode neural stimulation instrument and its desktop/web operator software; now developing FPGA-based acquisition and stimulation interfaces. Hands-on experience spanning Verilog, low-level firmware, PCB design, board bring-up, and automated test tooling.

Education

University of Alberta Expected 2026
B.Sc. in Electrical Engineering, Co-op *Edmonton, AB*

Technical Skills

Languages	C, C++, Python, Verilog, AVR/PIC/ARM assembly, MATLAB, Tcl, VBA
FPGA / SoC	Vivado, Vitis, PetaLinux, Quartus Prime, Gowin EDA; Zynq-7000, Cyclone IV, MAX II CPLDs, Tang Nano; RTL state machines, simulation/testbenches, timing constraints and report analysis, AXI DMA integration, JTAG, XSDB
Embedded	Timers, interrupts, DMA, ADC/DAC, UART, SPI, I2C; preemptive scheduling, synchronization, context switching; STM32, AVR, PIC, 8051-family MCUs (CH552, STC), Raspberry Pi
Hardware	Altium Designer, PCB layout, mixed-signal circuits, soldering/rework, board bring-up, ST-Link debugging; oscilloscopes, logic analyzers (sigrok / PulseView), VNAs, signal generators
Tools	Git, Linux, PySide6, Flask, Digilent WaveForms SDK, LTspice, PathWave ADS, Cadence Virtuoso, Ghidra, Binwalk, USBPcap, Wireshark

Engineering Experience

University of Alberta Research Lab 2026 – Present
Research Engineering Co-op — Hardware / Firmware Development *Edmonton, AB*

Rev-1: 48-electrode neural stimulation instrument

- Designed and brought up a four-layer PCB supporting eight independently routed stimulation lanes across 48 electrodes, with analog crosspoint switching, selectable return paths, and multiplexed current/voltage measurement.
- Developed biphasic, charge-balanced current-source/sink stages for 100–300 μA stimulation; validated operation across 3.3–22 kΩ loads on $\pm$7.8 V rails, with mean internal-sense tracking error below 0.7% in representative-load tests.
- Completed board-level fault isolation and rework across power, digital control, routing, and measurement circuits; verified electrode routing and eight-lane operation, resolving a shared-bus interaction to achieve less than 1.7% worst-case internal-sense error at 200 μA across tested loads.
- Built Python/PySide6 desktop and Raspberry Pi-hosted browser operators with timeline editing, hardware-timed waveform generation, route-aware scheduling, calibration, capture visualization, and safe-state controls.
- Implemented disk-backed long-capture recording and post-run transfer; retained 89.88 million paired samples in a 90-second, 1 MS/s mixed-topology stress test with zero lost/corrupt samples or discontinuities.

Rev-2: FPGA-based acquisition and stimulation development

- Developed Verilog SPI control and Intan-compatible emulator logic on Zynq-7000 and Cyclone IV platforms; used simulation testbenches and external board-to-board SPI tests to check protocol behavior and record integrity.
- Integrated timestamped acquisition, FIFOs, and AXI DMA into a Zynq prototype; debugged programmable-logic/ARM interfaces through Vitis test applications, UART diagnostics, and capture validation.
- Built and booted a custom PetaLinux image on a Zynq-7010 board, bringing up UART console access and Ethernet connectivity.
- Used Vivado and Quartus Prime for synthesis, implementation, pin/timing constraints, timing-report analysis, and JTAG programming; developed Tcl/Python scripts to support repeatable hardware builds and bring-up.

Engineering Experience — Continued

Seer Innovations Inc. Jan 2024 – Sep 2025
Engineering Support Specialist *Canada*

- Contributed firmware, test, and integration work for a cathodic protection rectifier replacement using controllable switch-mode supplies, measurement, logging, and remote monitoring.
- Produced inspection and corrosion-protection reports, including anomaly tables, engineering plots, data summaries, and supporting appendices.
- Automated survey-data processing and reporting with VBA and custom tools, including CSV-to-KMZ conversion for field mapping.

Weir Concepts Jun 2022 – Sep 2023
Shop Technician *Edmonton, AB*

- Operated CNC, plasma-cutting, and saw equipment; assembled and pressure-tested mechanical and hydraulic systems to API standards.

Selected Projects

Razer Mouse Control 2026
USB HID Reverse Engineering / Python Desktop Utilities

- Reverse-engineered DeathAdder 2013 and Viper Mini configuration protocols through native-library analysis, firmware extraction, and USBPcap/Wireshark capture validation.
- Built standalone Python utilities for DPI, polling, and device-specific lighting controls, with hardware readback verification and baseline restoration in reversible tests. github.com/trey109231/RazerMouseControl

4-bit Soft CPU on a CPLD 2026
Verilog / Quartus Prime — MAX II EPM240

- Implemented and hardware-validated a 4-bit accumulator CPU with a 12-instruction set, ROM program, conditional branches, and subroutine calls; generated bit-banged UART output through instruction-timed software.

Cathodic Protection Rectifier Controller 2025 – 2026
Capstone — Embedded / Firmware Lead

- Developed Particle Boron firmware for zero-cross-synchronized SCR phase control, RMS/DC measurement, calibration, and closed-loop voltage regulation on rectifier hardware.
- Built cloud telemetry, diagnostics, waveform capture, history logging, and a Flask web interface for system control and validation.

BadRTOS / PIC TinyRTOS 2026
Preemptive AVR RTOS and PIC16F84A Cooperative Runtime

- Designed and hardware-validated a preemptive ATmega328P RTOS in C/AVR assembly with manual context switching, separate task stacks, timer-driven scheduling, priorities, and task delays.
- Implemented semaphores, mutexes with priority inheritance, mailbox IPC, stack monitoring, and UART diagnostics on a microcontroller with 2 KB SRAM.
- Built a PIC16F84A assembly runtime with cooperative task timing, debounced inputs, software PWM, and bit-banged UART. github.com/trey109231/badRTOS

OpenAD3Flash 2026
Microcontroller Programming and Memory Access Tooling

- Built Python/WaveForms tools that use an Analog Discovery 3 for AVR ISP, PIC16 high-voltage programming, and EEPROM/SPI flash reads.
- Implemented bit-banged protocols, target-power control, signature detection, Intel HEX parsing, and flash/verify workflows. github.com/trey109231/OpenAD3Flash

LabLink — Legacy Instrument Automation 2026
Python / Streamlit / SCPI Instrument Drivers

- Built Python drivers and a unified Streamlit dashboard for an oscilloscope, waveform generator, and picoammeter across LAN/VXI-11, USB/VISA, and RS-232.
- Resolved legacy-instrument command sequencing and acquisition-state issues; implemented measurement logging, waveform export, replay scripts, and guarded output controls. github.com/trey109231/labLink